

# LivEdge: Haptic Live Stream Interaction on a Smartphone by Electro-Tactile Sensation Through the Edges

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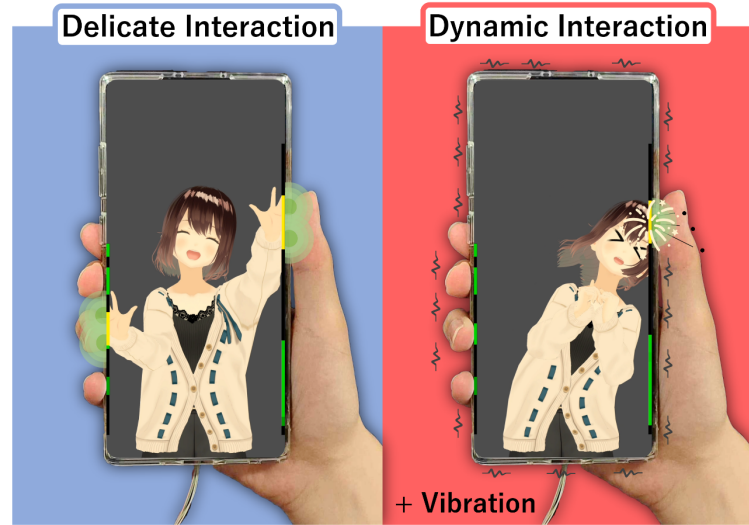


Figure 1: (Left) LivEdge presents spatial tactile information on both sides of a smartphone using electro-tactile stimulation. It enables viewers to interact with streamer on the screen through the edges of the smartphone. (Right) Delicate Interaction and Dynamic Interaction.

## ABSTRACT

We present LivEdge, a novel method for live stream interaction on smartphones utilizing electro-tactile sensation through the edges. Conventional interactions between users and a streamer on a smartphone are restricted to the streamer's response through user comments or effects. Our goal is to provide a more immersive interaction through the use of haptic technology. LivEdge can convey spatial tactile sensations through electrical stimulations from electrode arrays affixed to both edges of the smartphone. This spatial tactile stimulus represents the streamer's physical presence and movements in contact with the edge of the screen. Preliminary experiment showed LivEdge enhances the live stream experience.

## CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction(HCI); Interaction devices.**

## KEYWORDS

Mobile Haptics, Electro-Tactile Interface, Smartphone, Haptic Interaction, Live Stream

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## 1 INTRODUCTION

"Live streaming" refers to the real-time broadcasting of video content, primarily games and chats, to viewers on their smartphones or PCs. This type of digital live entertainment has acquired a significant number of new users in their leisure time during stay-at-home order due to the coronavirus pandemic, resulting in a rapid expansion of the market size. In addition, platforms for live streaming

utilizing avatars such as VTuber, gaining popularity in recent years, have also been introduced, providing significant value in terms of self-expression and creativity by transcending the physical limitations and one's appearance in the physical world.

The principal means of communication and interaction with viewers in live streaming, whether utilizing a physical body or an avatar, are commenting and gifting services. Some services enable viewers to participate in the streamed game indirectly or to convey non-verbal emotions to streamers [Yamazaki and Shirai 2021]. Although these services primarily comprise of visual and auditory entertainment, in the form of the streamer's appearance and effects on the screen, as well as the streamer's voice and musical sounds, there is a lack of strong interaction through physical contact, unlike in certain face-to-face events where fans are able to meet and shake hands with their favorite idols. We posit that such physical contact not only strengthens intimacy but also enhances the sense of immersion in the live streaming experience. In summary, remote haptic communication in live streaming is an underexplored area with significant potential.

In a prior research, There is an attempt to present electrical stimulation linearly at the edge of a smartphone [Groeger et al. 2019]. Building upon this concept, we propose LivEdge as a novel method of interaction in live streaming (Figure1 Left). LivEdge provides a spatial haptic sensation through electrical stimulation from an electrode array affixed to the edge of the smartphone. This sensation can emulate the body and movements of the streamer in contact with the edge of the screen, such as the streamer touching on the viewer's thumbpad throughout the live stream. Additionally, LivEdge can detect finger contact on the electrodes, providing feedback to the streamer on the position of the viewer's fingers.

As a prototype, we constructed a Live Stream system utilizing an avatar and integrated the proposed methodology.

## 2 SYSTEM

The overall system is composed of a webcam and a monitor for streamer (Figure2(a)), a streaming server (Figure2(b)), smartphones equipped with an electrical stimulation device that uses an electro-tactile display kit (Figure2(c)) [Kajimoto 2021]. The electrical stimulation device is powered by the smartphone via USB.

The system flow proceeds as follows: (1) Send streamer's video data captured by the webcam to the streaming server, (2) Utilize Webcam Motion Capture, a software for facial and motion tracking, to process the video data, (3) Generate electrical stimulation command data based on the tracking data in Unity, (4) Send the tracking data and electrical stimulation command data to the smartphone via WiFi, (5) Display an avatar on an Android application we have developed, and present tactile sensations. The intensity of the electrical stimulus can be easily changed on the application. In addition, the electrical stimulation device continually transmits contact information between electrodes and fingers to the streaming server, providing visual feedback on the monitor the streamer watches.

## 3 PRELIMINARY EXPERIMENT

We tested with three participants, all males, 21 to 27 years of age. To compare with a vibration of the smartphone, we evaluated the

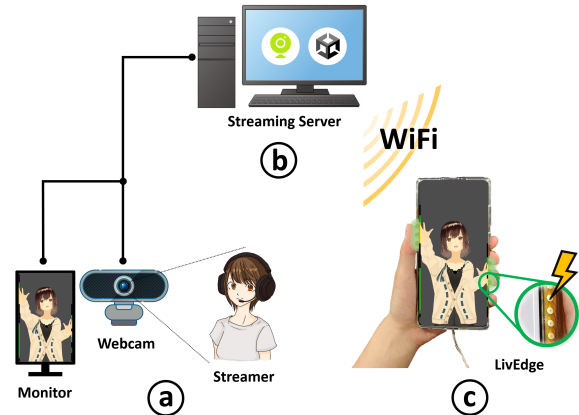


Figure 2: System Overview

user experience of vibration only, LivEdge only, and combination of both vibration and LivEdge. The vibration was regulated through the use of an Android API.

As a result, all participants exhibited favorable reactions towards the LivEdge, with specific feedback such as "I was taken aback by the realism of the sensation of the streamer's touch on my finger" and "I felt as though the streamer was present in my hand". In contrast, two participants reported that they deemed the vibration component to be unnecessary, citing reasons such as "the tactile sensitivity of the electrical stimulation was diminished" and "the vibration did not align with the visual representation". The other responded that either with or without vibration would be fine, and also noted that vibration was well-suited for more vigorous movements such as slapping and headbutting.

## 4 EXPERIENCE

We developed delicate interaction and dynamic interaction (Figure1 Right) in live streaming. As a delicate interaction, slow movements such as stroking and soft touch are presented by LivEdge. Slow and soft tactile sensations will relax and intimate viewers. As a dynamic interaction, vigorous movements such as slapping and headbutting are presented with vibration additionally. Combined with delicate interactions, viewers will enjoy a well-balanced interaction.

## ACKNOWLEDGMENTS

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